

140th Master's Thesis

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A Scalable Bridge Health Monitoring System using an IoT Sensor and Cloud Computing

February 2024

**The Graduate School
Chung-Ang University
Department of Civil Engineering
Major In Structural Engineering
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**Presented to the Faculties of the Chung-Ang University in
Partial Fulfillment of the Requirement of the
Master's Degree**

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by

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Civil Engineering in the Graduate School
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2024

Abstract

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Bridges are critical infrastructure that must be regularly maintained to ensure their safety and reliability. However, it is expected that the percentage of aging bridges in South Korea, with a service life of over 30 years, will rapidly increase from 17.9% in 2020 to 49.7% by 2030. The conventional inspection methods, which heavily rely on manpower, face limitations in managing the growing number of aging bridges due to a decline in construction workforce. Moreover, during safety evaluations assessing the load carrying capacity of bridges, which indicates their resistance to dynamic loads, displacement must be measured. However, traditional displacement measurement methods (LDV, LVDT) are impractical for bridges over rivers or with tall bridges, making proper inspections challenging.

To overcome these issues, this paper proposes the development of an IoT sensor system (JANET) and a cloud-based bridge monitoring system. JANET can simultaneously measure three-axis acceleration and up to 8 channels of strain. It maintains a low-power state using an EDS-based trigger circuit and acquires data when triggered by excessive vibrations or at scheduled intervals. Additionally, it embeds an LTE Cat 1 modem to

wirelessly transmit the collected data to the cloud server. The cloud server consists of node.js scripts for data acquisition and python scripts for data processing. It includes algorithms for strain offset calibration, acceleration-strain-based displacement estimation, calculating key indices of the bridge in real-time from the measurement data. The resulting data is stored in databases like InfluxDB and MySQL, and these databases are integrated with web-based data visualization dashboard.

The developed system has been installed at the southern end of Seongsan Bridge in Seoul since July 2023. During the measurement period, it has been confirmed that the maximum displacement (2.61 mm) and maximum crack width (66.8 μm) observed at the sensor attachment point are stable, deviating by 6.94% and 22% from the design criteria, respectively.

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1 Introduction

1.1 Backgrounds

In South Korea, the majority of infrastructure facilities were constructed during the economic boom of the 1970s and 1980s [1]. This historical development, while contributing to rapid economic growth, has given rise to a significant challenge; aging infrastructure [2]. Currently, the aging rate of bridges, a critical component of the infrastructure network, is at 17.9% as of 2020. Projections indicate a concerning escalation to 49.7% within the next decade [3]. This trend is not limited to bridges alone; other vital structures such as ports, rivers, and tunnels are also anticipated to experience a notable increase in their aging rates.

The consequences of this aging infrastructure are evident in recent incidents involving major bridges [4-6]. Notable examples include the identification of issues in the Cheongdam 1 Bridge in 2016 and the Seongsan Bridge in 2021, necessitating extensive and costly maintenance [7]. The collapse of the Jungja Bridge in Seongnam, classified as a Class 3 facility, and the Geumgo Bridge in the current year further emphasize the risks associated with aging infrastructure [8]. The implications extend beyond immediate safety concerns, encompassing substantial financial burdens due to the high costs associated with repairs and maintenance.

Bridges are indispensable elements that facilitate the connection of people and goods. Their collapse not only poses a direct threat to lives but also results in significant economic repercussions, including elevated repair costs and disruptions to the smooth flow of goods and services. Consequently, addressing the challenges posed by the aging infrastructure is paramount to ensuring the safety and economic stability of the nation.

In light of these issues, there is an urgent need for strategic interventions and policy measures. These may include the development of comprehensive plans for infrastructure renewal, the establishment of regular inspection and maintenance protocols, and investments in innovative technologies to enhance the longevity of structures. A proactive approach to tackle the aging of various infrastructure components, extending beyond bridges, is crucial for the overall resilience and sustainability of the nation's infrastructure.

In summary, the aging of domestic infrastructure in South Korea, particularly in the case of bridges, demands immediate attention. This section serves as an introduction to the challenges posed by aging infrastructure, laying the groundwork for subsequent discussions on policy implications and potential future directions to address this pressing issue.

1.2 Literature Review

1.2.1 Indirect Displacement Estimation

Displacement serves as a key indicator of a bridge's condition and is measured for the evaluation of its dynamic resistance capacity during the evaluation of the load carrying capacity of a bridge [9]. Typically, displacement is measured directly using displacement meters such as Laser Doppler Velocimeter (LDV) or Linear Variable Displacement Transducer (LVDT). However, these methods often pose challenges, requiring ground reference points, and prove difficult for displacement measurement in cases of bridges that are hard to access, such as long-span or river bridges.

Recognizing the challenges associated with direct measurements, researchers have shifted their focus to indirect displacement estimation methods [10-12]. Within the category of indirect methods, various approaches have been explored. Some researchers propose estimating displacement by measuring acceleration responses, while others investigate converting strain-based measurements into displacement estimates. While these indirect methods offer alternatives, the single measurement-based displacement estimation techniques have limitations. For example, there are issues such as low-frequency drift in acceleration-based methods and challenges related to the neutral axis in strain-based measurements [13].

To enhance the accuracy and reliability of displacement estimation, researchers have turned to data fusion techniques. Notably, Park et al. [14]

introduced a data fusion approach that combines strain and acceleration measurements to derive pseudo-static displacement. This method requires a minimum of three strain sensors and one acceleration response for precise estimation. The method also has been validated for accuracy through multiple on-site experiments, and it can be effectively applied in the field when acceleration and multi-channel strain measurements can be acquired simultaneously [15-17].

1.2.2 Structural Health Monitoring Sensor

In the field of Structural Health Monitoring (SHM), extensive research has been conducted to explore various sensors that can enhance the effectiveness of monitoring systems. Over the past decade, there has been a notable shift from traditional wired sensors to wireless sensor technologies [18]. These wireless sensors offer advantages such as ease of installation, cost-efficiency, and streamlined data processing.

In the realm of SHM in the last ten years (i.e., 2012~2022), four notable wireless sensors have been deployed, namely Martlet, Xnode, Waspote, and G-link-200 [19]. Each sensor has its unique characteristics and capabilities. For instance, Marlet, developed at the University of California, San Diego, is a sensor that excels in basic monitoring (accelerometer, temperature, humidity) within a 100-meter range. However, data resolution and functionality are limited. Xnode, developed at the University of Illinois at Urbana-Champaign, is a powerful sensor that boasts high processing, diverse sensor compatibility, and 24-bit resolutions. It tackles complex monitoring across 1km, but at the cost of increased power consumption and deployment complexities. Waspote, developed at the University of Twente, is a small and portable sensor that shines in long-term, low-cost monitoring for diverse parameters (temperature, humidity, voltage, etc.) within 200 meters. However, data resolution is limited, and it is vulnerable to harsh environments. Lastly, G-link-200, developed at the

University of Stuttgart, is a sensor that specializes in three-axis acceleration with high-speed data transmission across 1km. It excels in specific applications, but its limited functionality restricts its versatility.

Overall, the continuous exploration and development of wireless sensor technologies in SHM demonstrate a commitment to overcoming challenges associated with traditional wired sensor systems, paving the way for more efficient and advanced structural health monitoring solutions.

1.3 Objectives and Thesis Overview

The study aims to develop a system for long-term monitoring of a real-site bridge for health assessment. The proposed system consists of an IoT sensor and cloud-based server, providing users with convenient bridge management capabilities. The proposed system is applied to a real-site bridge, named Seongsan bridge in Seoul, and monitored for 4 months.

The highlights of contribution and main objectives are as follows.

- Development of an IoT sensor for simultaneous measurement of acceleration and multi-channel strain
- Development of a cloud-based computing server with incorporation of databases and cloud-based operations
- Application of the system to Seongsan Bridge, Seoul for 4 months
- Long-term data analysis of a real bridge response and web-based dashboard integrated with databases for data visualization.

The following chapter is composed as of 5 chapters. The chapter 2 discusses the proposed design for JANET. The chapter 3 presents the proposed framework of cloud-computing server for data management. The chapter 4 shows the application of the proposed system to the Seongsan Bridge, Seoul for 4 months and its discussion. Finally, the thesis concludes with a summary and outlines future work in chapter 5.

1.4 Overview of the Proposed System

This study aims to develop a data-driven bridge management system for long-term bridge health monitoring. The overall scheme of the proposed system is presented in Figure 1-1, which is subdivided into two parts. The first part involves the development of the JANET, which is a smart sensor that can measure 3-axis acceleration and up to 8 channels of analog sensor (strain gauge, crack meter, etc.) simultaneously. To save its power, two types of event-driven triggering function (time-based and vibration-based) is applied to the design of the circuit. LTE modem can be connected to the JANET for data transmission.

The second part is the cloud-based server that enables wireless data acquisition from the sensor. Server stores acquired data on databases and apply embedded cloud computing operations to conduct data analysis. Processed data in database is visualized on web-based dashboard and display bridge's data-driven real-time assessment. To expand the proposed system, serverless architecture and scalability in cloud-computing are considered when the data schema is designed.

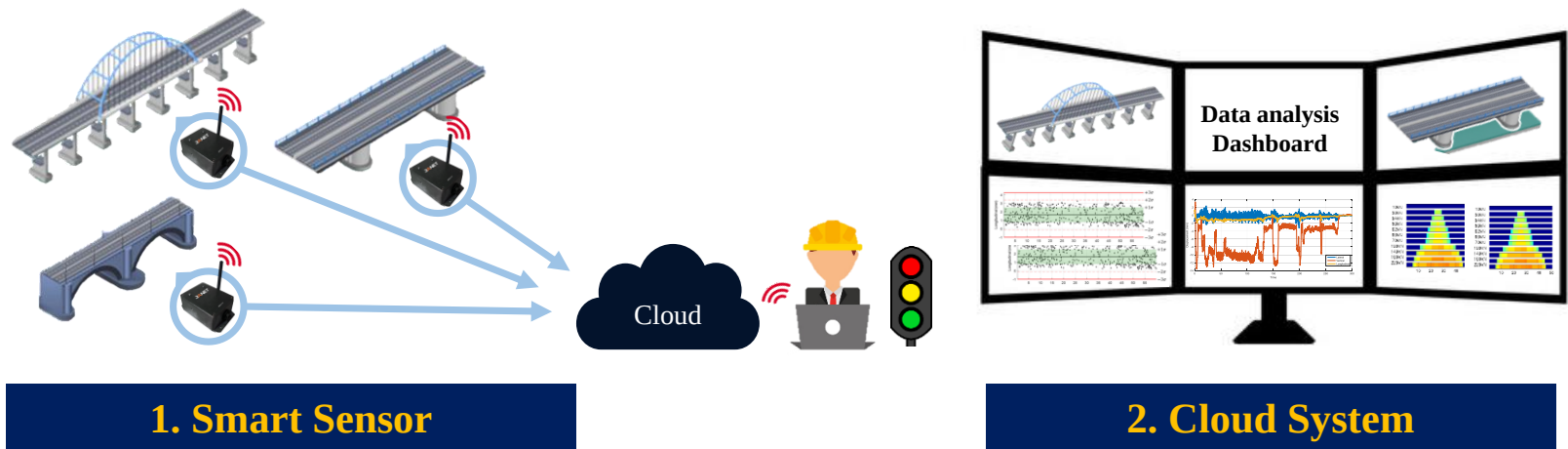


Figure 1-1: Overview of the proposed system

2 Proposed Sensor Development

This chapter introduces JANET, a sensor developed for long-term bridge monitoring. JANET incorporates a Microcontroller Unit (MCU) for rapid processing, an Analog-to-Digital Converter (ADC) for simultaneous high-resolution multi-channel data measurement, and an LTE modem for wireless transmission of measured data to the server. Additionally, it is equipped with an Event Driven System (EDS) circuit and a solar charging system, enabling it to operate semi-permanently in the field without the need for battery replacement.

2.1 System Design

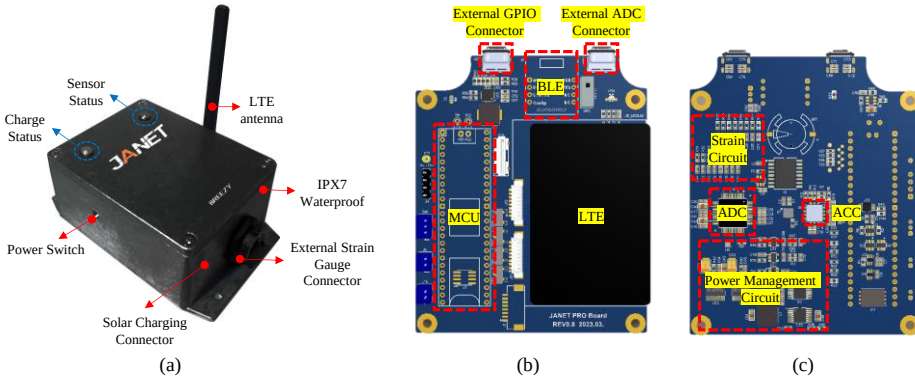


Figure 2-1: Wireless sensor JANET: (a) packaged (b) front side (c) back side

In contrast to previous wireless bridge monitoring sensors like Imote2, which faced development limitations due to production suspensions [20], JANET addresses these issues by adopting an Arduino-compatible MCU; an open-source hardware. The circuit diagram, printed circuit board,

and hardware description language are all open to the public, ensuring sustainable development. Two key considerations for MCU selection were a clock speed for wireless data transmission and RAM size for multi-channel data acquisition with high-sampling rate.

Table 2-1: Comparison of Micro Controller Unit

	Teensy 4.1	Arduino Due	Arduino Mega	Adafruit Metro M5	Sparkfun ESP32 Thing
Core	ARM Cortex-M7	ARM Cortex-M3	ATmega2560	ARM Cortex-M4	Dual core Tensilica LX6
Clock Rate	600 MHz	84 MHz	16 MHz	120 MHz	240 MHz
RAM	9 MB	96 KB	8 KB	192 KB	520 KB
SDIO	Micro SD	-	-	-	-
Serial Ports	8	4	4	1	3
SPI Ports	3	1	1	1	3
I2C Ports	3	-	-	1	2

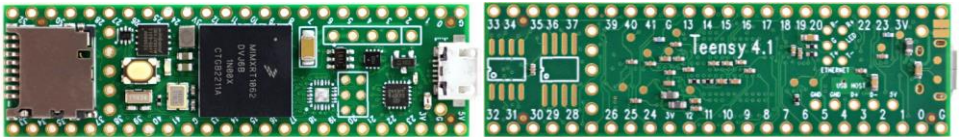


Figure 2-2: PJRC Teensy 4.1 microcontroller

Among various Arduino-based MCUs, Teensy 4.1 meets the requirements, boasting high specifications (refer to Table 2-1). Teensy 4.1 offers ample memory with 7936kB and 1024kB in flash and RAM, respectively, along with an additional 8MB PSRAM for extra data storage [21]. It features a 32-bit ARM Cortex-M7 core with Arduino IDE support,

operating at a clock speed of 600 MHz, providing ample processing speed.

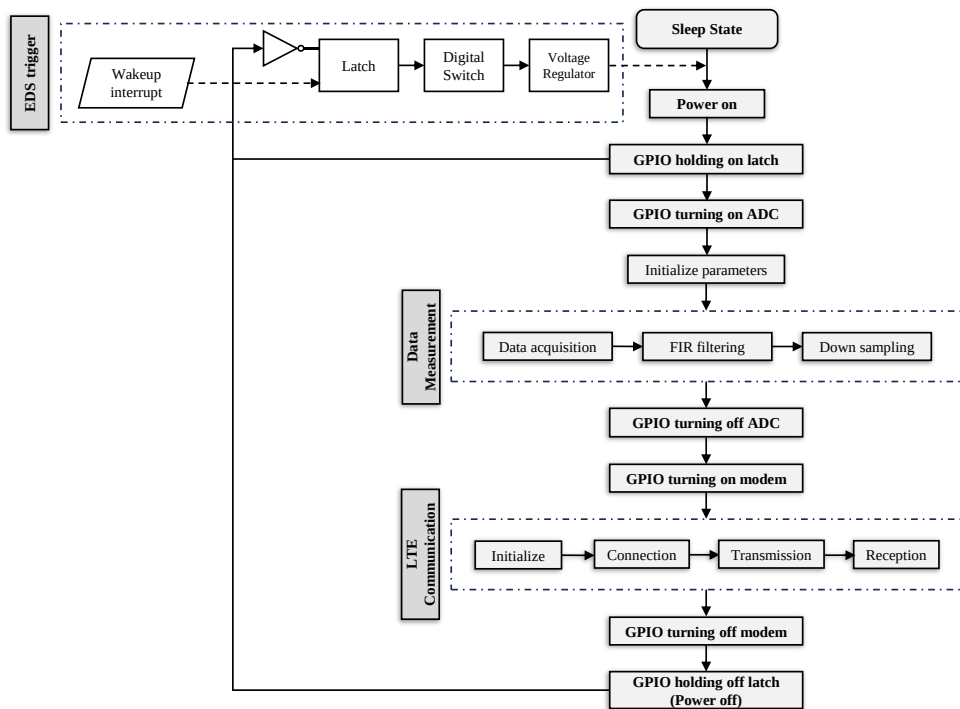


Figure 2-3: JANET software flowchart

Figure 2-3 illustrates the software flow diagram running on JANET. In this study, JANET is designed to wake up the sensor at specific times to acquire data, rather than being an always-on system that measures data 24 hours a day. GPIO control related to the latch at the beginning and end of the flowchart is associated with this, and further details will be discussed in Chapter 2.2.

After the MCU is turned on, it activates the ADC, initializes sensing parameters, and proceeds with the measurement. The sensor can simultaneously measure 3-axis acceleration and multi-channel analog

sensors through the built-in accelerometer and ADC. Upon completing the measurement, the MCU utilizes the LTE modem connected to JANET to transmit the measured data to the cloud server.

JANET can also be equipped with a Bluetooth module, and through the external GPIO connector presented in Figure 2-1, additional modules can be attached to the case exterior to perform various functions beyond the basic settings provided. This versatility allows JANET to be applied seamlessly to any site or facility aiming to measure data for an extended period and collect measurement data wirelessly. Detailed tasks performed on JANET will be introduced sequentially in the subsequent sections.

2.2 Event-Driven based Power Management Circuit

The occurrence of sudden events in structures is unpredictable; hence, the monitoring system must be continuously in an "on" state. However, if the sensor is always on, measuring data 24 hours a day, it would deplete its battery quickly and cause data inundation. Most wireless sensors for infrastructure monitoring are deployed in difficult-to-access areas, making battery replacement inconvenient and expensive. To address this challenge, EDS is a promising solution [22-23]. EDS only wakes up the sensor when an event occurs. The event is defined in two trigger cases.

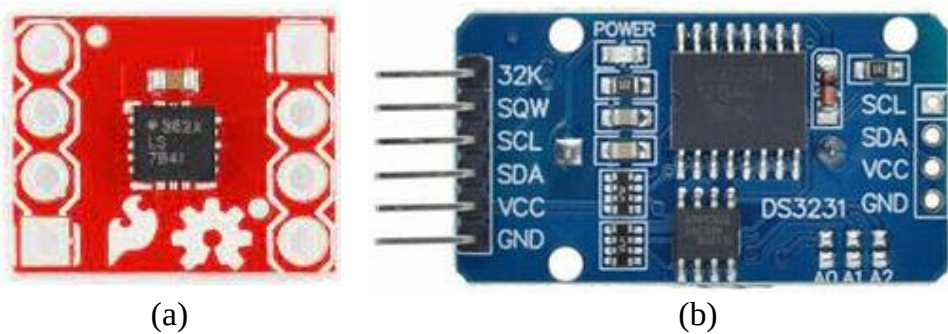


Figure 2-4: Trigger modules: (a) ADXL362 (b) DS3231

One type of event trigger in EDS is a vibration-based trigger typically operated by an accelerometer. When the measured acceleration level exceeds the preset threshold, the accelerometer is triggered, generating an interrupt signal. JANET's EDS utilizes the ADXL362, a three-axis microelectromechanical system (MEMS) accelerometer developed by Analog Devices, for its vibration-based trigger [24]. This choice is

attributed to several advantages offered by the ADXL362: (1) The ADXL362 operates at low power consumption of 13 μA in low-power noise mode, essential for battery-powered EDS systems. (2) With high precision at 1 mg, it reliably detects subtle changes in vibration for accurate event detection. (3) Its fast-sampling rate of 100 Hz ensures a prompt response to significant vibration events, facilitating timely data collection. These features make the ADXL362 a well-suited choice for vibration-based trigger systems, contributing to the effectiveness and efficiency of JANET's EDS.

Another type of event trigger in EDS is a timer-based trigger, waking up the sensor at regular intervals, even in the absence of detected events. This is useful for applications where regular data collection is important, such as environmental monitoring. For the timer-based trigger of JANET's EDS, DS3231, a Real-Time Clock (RTC) developed by Maxim Integrated, is employed [25]. DS3231 has low current consumption of 200 μA and can generate an interrupt signal when a user-defined time is met.

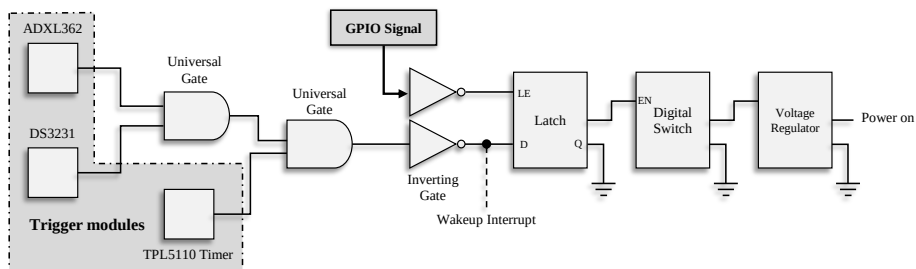


Figure 2-5: Functional block diagram of EDS circuit

JANET separates the power supply within the circuit board, ensuring that power from the battery is not always distributed to all circuits. Instead, only the EDS circuit receives power and remains in standby (inactive mode). Power is supplied to the entire circuit only when interrupt signals occur from ADXL362 or DS3231. For this purpose, a latch and digital switch have been implemented.

The latch consists of two inputs and one output: latch enable (LE), D, and Q, respectively. When LE is high, Q follows D, and when LE is low, Q retains the previous state of D [26]. Initially, LE is pulled up to allow the trigger signal to pass through, and Q is low because there is no trigger signal (D is low). Upon the generation of the trigger signal, combined through two universal gates, the resulting signal becomes low (active low). After passing through an inverting gate (i.e., wakeup interrupt), D goes high, causing Q to go high as well. This action activates the digital switch, turning on the MCU and other modules.

However, the trigger signal is reset after a few seconds to capture the next event. If the wakeup interrupt turns low when LE is high, the MCU is turned off during its operation. To address this issue, the MCU sends a high signal through its digital pin (General Purpose Input/Output; GPIO), which, when inverted, becomes low and enters LE, maintaining its power supply. Once the entire process is completed, the MCU sends the GPIO signal low (i.e., LE becomes high), returning JANET to inactive mode.

2.3 Design of Circuit for Simultaneous Acceleration and Strain Measurement

Conventional wireless sensor architectures, which used multiple leaf nodes and a single gateway node, suffered from time synchronization issues and data loss due to complex network configurations [18]. Additionally, managing the power of each sensor individually made them economically inefficient. To address these challenges, JANET was designed as a single-node architecture capable of acquiring 3-axis acceleration and up to 8 channels of analog sensor data (strain gauge, displacement sensor, etc.) simultaneously, eliminating the need for time synchronization and minimizing data loss.

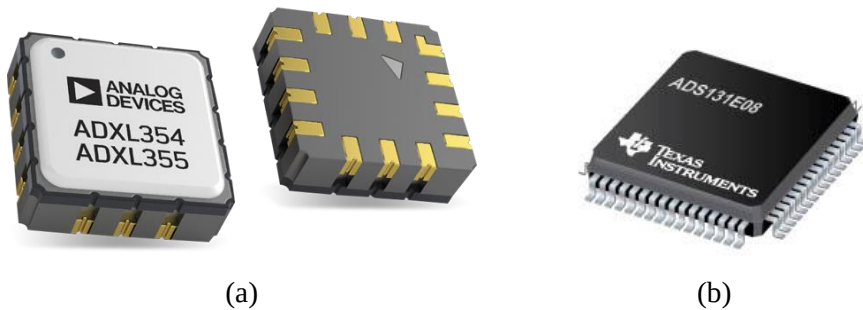


Figure 2-6: Accelerometer and ADC: (a) ADXL355 (b) ADS131E08

JANET leverages the ADXL355 digital accelerometer developed by Analog Devices for high-accuracy and precision measurements. The ADXL355's low-noise density ($25 \mu\text{g}/\sqrt{\text{Hz}}$), minimal 0 g offset drift, temperature-stable offset drive, selectable sensing ranges ($\pm 2\text{g}$, $\pm 4\text{g}$, $\pm 8\text{g}$), and measurement frequency of up to 4 kHz enable JANET to perform

accurate measurements in various environments and applications. JANET also utilizes the ADS131E08, a 24-bit Analog Digital Converter (ADC) developed by Analog Devices [27], to enable simultaneous measurement of multi-channel analog sensors. The 24-bit ADC supports simultaneous sampling of up to 8 channels, achieving low power consumption of 2mW per channel. It supports a maximum sampling rate of 64 ksp/s and a programmable gain amplifier (PGA) with a wide range of gain factors (1, 2, 4, 8, 12), making it suitable for low-level signals.

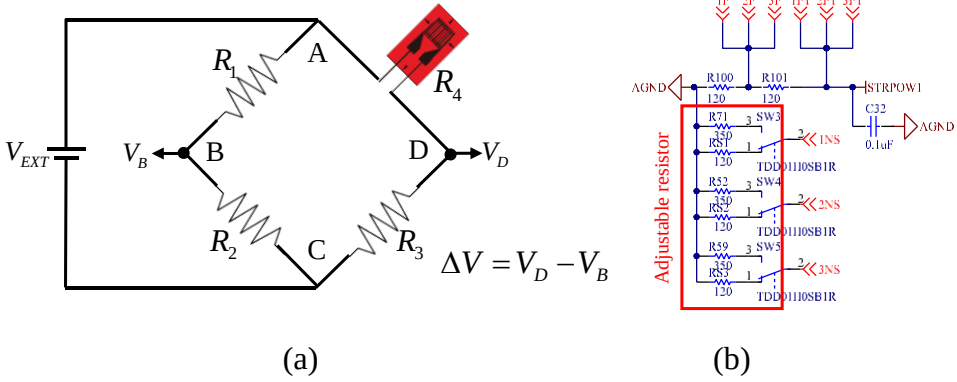


Figure 2-7: Sensor circuitry: (a) wheatstone bridge (b) adjustable resistor switch

$$\Delta V = \left[\frac{-\Delta R}{4R + 2\Delta R} \right] V_{EXT} \quad (1)$$

$$\Delta V = -\frac{GF \cdot \varepsilon}{4} \left[\frac{1}{1 + \frac{GF \cdot \varepsilon}{2}} \right] V_{EXT} \approx -\frac{GF \cdot \varepsilon}{4} V_{EXT} \quad (2)$$

A Wheatstone bridge was designed to enable accurate strain measurement in Figure 2-7 (a). The bridge circuit consisted of four resistors, R_1 , R_2 , R_3 , and R_4 . For a quarter bridge, $R_1 = R_2 = R_3 = R$ and $R_4 =$

$R + \Delta R$ is only active resistance. Then, the differential voltage ΔV is calculated as:

$$\varepsilon = \frac{\Delta V \cdot 4}{GF \cdot V_{EXT}} \quad (3)$$

where $GF = (\Delta R/R)/\varepsilon$ is a gauge factor of strain gauge, V_{EXT} is excitation voltage and ε is strain. Then, strain value is obtained as:

To deliver stable excitation voltage, JANET employs the TPS7A20-18PDBVR [28], a linear regulator with a 300 mA current capacity, which sets V_{EXT} to 1.84 V. To improve field applicability, R3 was designed to be selectable to balance R4. So, users can choose the internal 120 Ω or 350 Ω resistor to match the resistance of the strain gauge they are using or can configure a half bridge without using R3 if necessary.

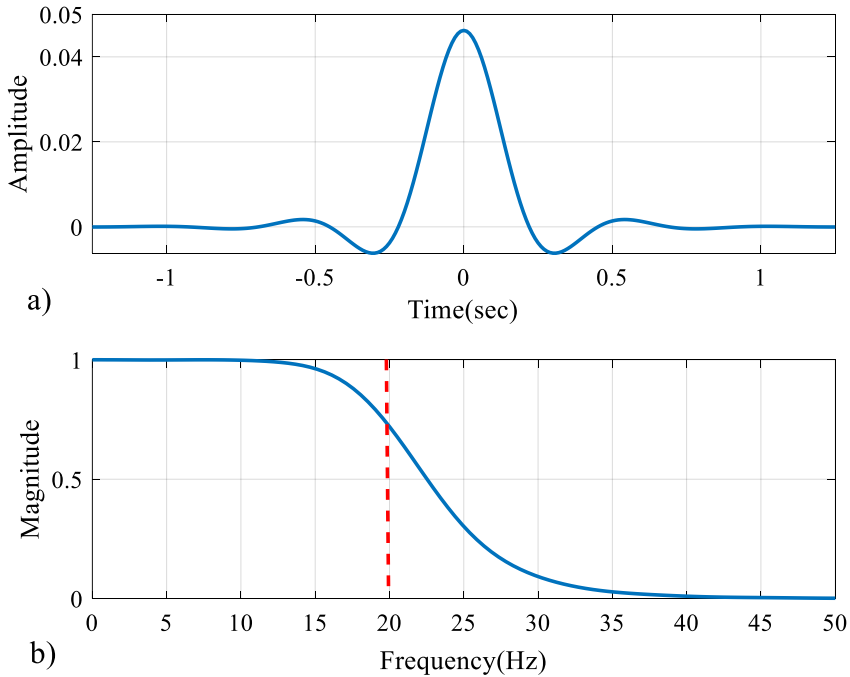


Figure 2-8: Designed FIR filter: (a) filter coefficients (b) frequency response

JANET employs real-time software-based anti-aliasing and a Finite Impulse Response (FIR) low-pass filter on the measured raw data at 1000Hz, attributed to linear phase delay, stability, and ease of implementation. The transfer function of the filter is represented as:

$$H_L(f) = \frac{1}{1 + (\frac{2\pi f}{\beta_L})^{2n}} \quad (4)$$

where β_L is the regularization factor, f is the frequency and n is the order of the filter. The regularization factor β_L can be expressed as:

$$\beta_L = \sqrt[n]{\frac{\alpha_L}{1 - \alpha_L}} 2\pi f_c \quad (5)$$

$$\alpha_L = H_L(f_c) = \frac{\beta_L^{2n}}{(2\pi f_c)^{2n} + \beta_L^{2n}} \quad (6)$$

where α_L represents the accuracy factor for the magnitude of the transfer function at cut-off frequency (f_c). The f_c is located at the edge of the passband as α_L approaches one. The FIR filter coefficients can be computed through inverse discrete Fourier transforms as:

$$C_L(p) = \frac{2}{f_s} f_c \int_0^{1/2rT} \frac{\beta^{2n}}{(2\beta f_c \tilde{f})^{2n} + \beta^{2n}} \cos(2\pi p r_c \tilde{f}) d\tilde{f} \quad (7)$$

where $r_c = f_c/f_s$ that is cut-off to sampling frequency ratio, and $\tilde{f} = f/f_c$ is the normalized frequency. The deployed FIR filter has f_c , α_L , and n of 20Hz, 0.99 and 7 respectively. The f_c threshold is set as 20Hz to avoid unnecessary noise as shown in Figure 2-8 (b). A total of 275 coefficient

samples were used for the FIR filter. Figure 2-8 (a) shows the coefficients and transfer function of the FIR filter.

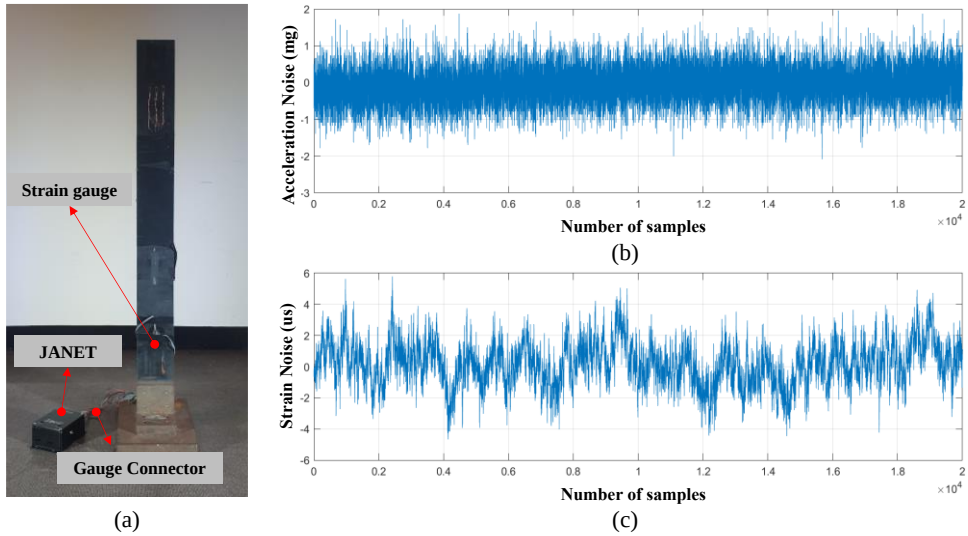


Figure 2-9: Lab-scale experimental setup and noise levels: (a) setup (b) acceleration noise (c) strain noise

Further, noise analysis is performed over 60 seconds in a vibration-free environment and shown in Figure 2-9 to find a minimum a sensor can measure, representing an effective number of bits. The Root Mean Square (RMS) for the acceleration noise graph is 0.1mg and for strain, the RMS is 1.5 μ s. The sensitivity of the accelerometer is 0.4 mg/V $\pm$ 2g sensing range [29].

2.4 LTE Data Communication

To automate data collection, LTE modems were implemented in JANET. Various types of IoT communication, such as LoRa, LTE Cat 1, NB-IoT, and EC-GSM-IoT, exist. However, considering factors such as the ability to quickly transmit a large amount of measurement data, development convenience, and modem cost, the LM6 modem developed by M2M in Korea was adopted as a communication module [30].



Figure 2-10: LTE Cat 1 modem: LM6

The LM6 is an LTE Cat 1 communication modem operating at 5V with an RS232 serial interface. It consumes 60mA in the idle state and up to 800mA at maximum. The modem measures 40(W)×52(H)×9(T), making it compact and convenient for sensor packaging. It is also easy to develop, providing all modem functions through AT commands. The LM6 supports various communication protocols such as TCP/IP, FTP, HTTP(s), SSL, and PPP. In this study, a data transmission system using the Message Queuing Telemetry Transport (MQTT) method, a lightweight messaging protocol

based on many TCP/IP protocols, was adopted.

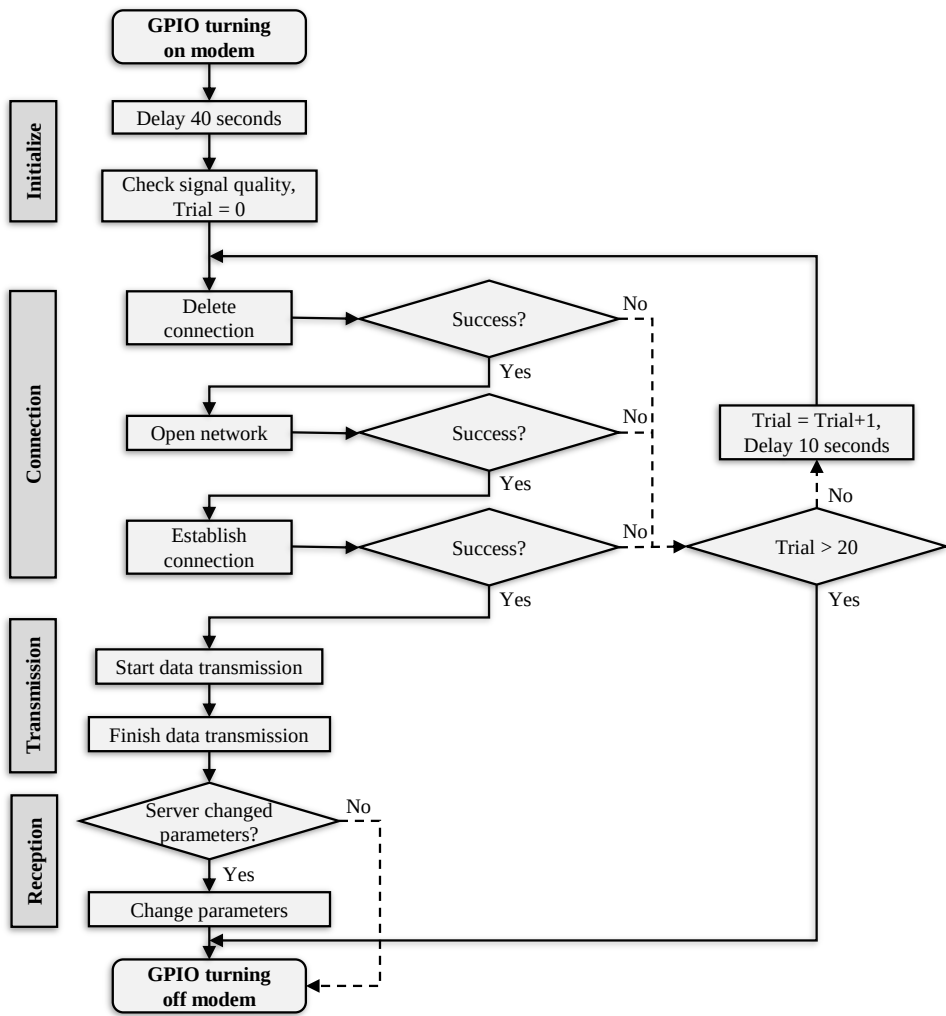


Figure 2-11: Wireless communication flowchart

After data acquisition is complete, the MCU activates the digital switch through the GPIO to energize the LM6. This conserves power, as data is not transmitted during acquisition. Once powered and situated on a service network, LM6 autonomously establishes an internet connection.

This involves a 40-second waiting period for internet access confirmation by JANET. Upon successful internet access, LM6 connects to the cloud server's MQTT broker. If any connection task fails, LM6 retries up to 20 times. The success of the AT command is confirmed through the acknowledgment (ACK) response. Once successfully connected, LM6 transmits the collected 60,000 data points. These data are packaged in groups of five points for transmission. Following transmission, LM6 waits for 10 seconds to receive and implement any measurement setting changes commanded by the server. Upon completion of all tasks, the MCU releases the latch pin and returns to inactive mode.

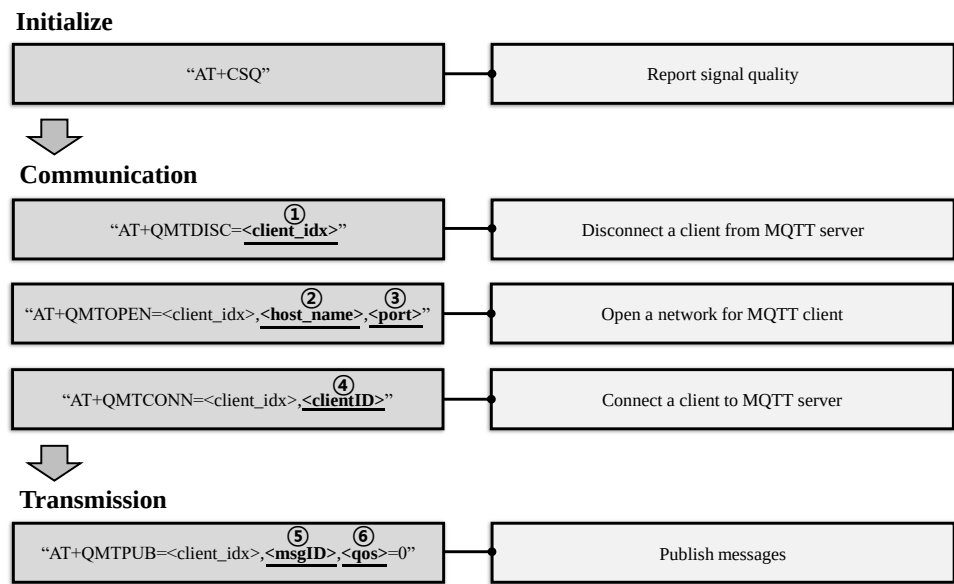


Figure 2-12: MQTT commands for LM6

- <client_idx>: The unique identifier of the MQTT client (e.g., "0")

- <host_name>: The hostname of the MQTT broker (e.g., “192.168.0.1”)
- <port>: The port number of the MQTT broker (e.g., “1883”)
- <clientID >: The ID of the MQTT client (e.g., “seongsanSouth1”)
- <msgID>: The ID of the MQTT message. Values from 0 to 65535 can be used (e.g., “0”)
- <qos>: The Quality of Service (QoS) of the MQTT message (e.g., “1”)

2.5 Power Consumption Analysis

The power consumption of JANET was investigated. JANET typically operates in an inactive mode, awaiting an event to occur. As mentioned in Chapter 2.2, only the EDS-related circuits are powered in this mode. The current consumption of JANET in inactive mode ($I_{inactive}$) can be calculated from the devices related to the EDS circuit (see Table 2-2) [31-36].

Table 2-2: Current consumption of JANET in inactive mode

Device	Current consumption
ADXL362 (Accelerometer)	1.8 μ A
SN74AUP1G58 (Universal gate)	4.5 μ A
ZS-042 (DS3231)	200 μ A
TPL5110 (Power timer)	35 nA
MIC94064 (Voltage regulators)	2 μ A
SN74LVC1G373 (D-type latch)	10 μ A
TPS75233-EP (Digital switch)	125 μ A
MAX8903AETI+T (Charger)	100 μ A
Total	443.3 μA ($I_{inactive}$)

When a wakeup interrupt occurs, JANET switches to active mode, measures data for 10 minutes following the flowchart shown in Figure 2-3, and transmits the measured data to the cloud server via LTE. Using a DC power supply, the current consumption for each state is depicted in Figure 2-13.

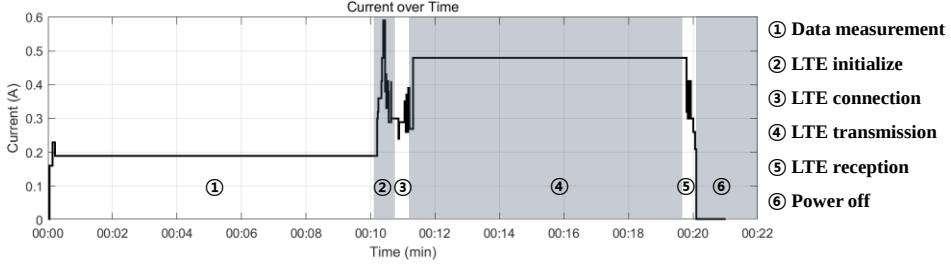


Figure 2-13: Current consumption of JANET in active mode

According to equation 8, I_{Active} is calculated to be $302.2mA$, where $T_{Active} = 0.33 \text{ hours}$ (20.1 min). The current consumption of the sensor per day depends on the number of wakeup interrupts that occur. In this study, a timer-based trigger is set up to occur every hour, and a vibration-based trigger to occur up to 5 times a day if the vibration exceeds a threshold of 120 mg.

$$I_{Active} = \frac{\int I dt}{T_{Active}} \quad (8)$$

Thus, the total operation time T_s is estimated as:

$$T_s = (N_{rtc} + N_{vib})T_{Active} \quad (9)$$

where N_{rtc} , N_{vib} are the number of time-based triggers and the number of vibration-based triggers, respectively (i.e., $N_{rtc} = 24$, $N_{vib} = 5$), resulting in $T_s = 9.57 \text{ hours}$. This leads to the current consumption of JANET per day (I_{day}).

$$I_{day} = I_{Active}(T_s) + I_{Inactive}(24 - T_s) \quad (10)$$

A total of three 18650 lithium-ion rechargeable batteries were deployed to have the battery capacity up to 10,500 mAh. Assuming no

external charging, the service life of the sensor can be calculated as follows:

$$Service_{Life} = \frac{Capacity(mAh)}{I_{day}} \times 0.8 \quad (11)$$

where 0.8 is a compensation factor, addressing external and environmental conditions that affect battery life [37]. In result, $Service_{Life}$ is calculated to be 2.89 days, which is insufficient to conduct long-term monitoring in the field.

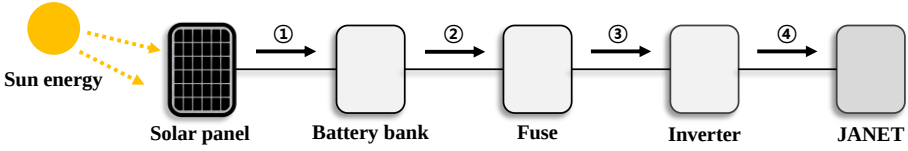


Figure 2-14: Solar-powered charging system diagram

Therefore, a charging system was applied to charge the lithium battery inside the sensor through a 60W solar panel. However, if the solar panel is connected directly to the sensor's internal battery, the sensor will be discharged if solar charging is not possible for more than three days. To prevent this, the solar panel herein is configured to charge the sensor's lithium battery through a 60Ah battery bank. This allows the sensor to be charged smoothly for up to 20 days even if the weather is not good. In addition, to prevent sensor damage caused by lightning strikes or unexpected overcurrent, a reverse current protection function to prevent overcharging was supported.

3 Proposed Cloud-Computing Framework

This chapter describes the configuration of the cloud server and the general framework for databases. The system is divided into several parts, as shown in Figure 3-1. Each part is described in the subsequent sections according to the cloud computing operation.

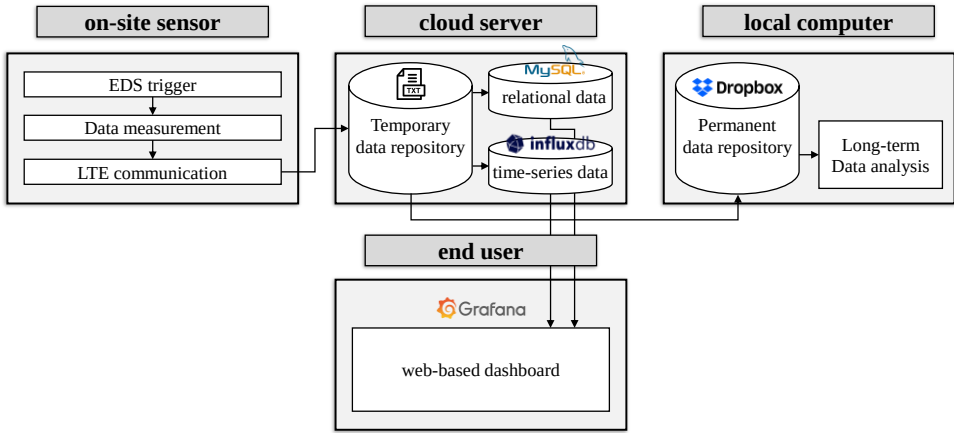


Figure 3-1: Proposed cloud computing framework

3.1 Overview

A cloud server based on Amazon EC2 was configured to receive measured data from JANET. In this study, a c5d.xlarge instance was used. c5d.xlarge offers 4 virtual CPUs, 8GB of memory, and data storage using EBS. It also supports high-performance networking, making it suitable for applications and database workloads that require high data transfer speeds and computing performance.

Table 3-1: Detailed specification of selected EC2 instance c5d.xlarge

Specification	c5d.xlarge
vCPUs	4
Memory	8GB
EBS bandwidth	Up to 4,750 Mbps
Networking	Up to 10 Gbps
Storage	EBS
Processor	2 nd generation Intel Xeon Scalable Processors
Turbo frequency	3.6 GHz
Single-core turbo frequency	3.9 GHz
Price	\$0.075 per hour

Two types of scripts are running on this EC2 server: a Node.js-based script for receiving data and a Python-based script for analyzing data. First, the Node.js-based script runs an MQTT broker and subscribes to a pre-configured topic. JANET connects to the server via an LTE modem and publishes the measured data to the same topic. This causes the measured data from the sensor to be transmitted to the server. If no new data is received within 20 seconds after the latest data is published, the Node.js script considers the data transmission from the sensor to be complete and temporarily stores the data received so far in a text file. It then changes the setting value related to the measurement setting of the sensor if necessary. This allows users to remotely control the measurement settings of sensors attached to bridges without having to visit the site.

Since the Node.js-based script and the Python-based script exist in the same cloud, the Node.js script, after receiving data, triggers the Python-based script to notify the data to be analyzed by sending the file name of the received data through Flask. Flask is a micro web framework based on Python that can be used to create concise web programs and connect to various modules. The Python script opens the target file, performs built-in operations, and uploads the processed data to the database and finishes its task.

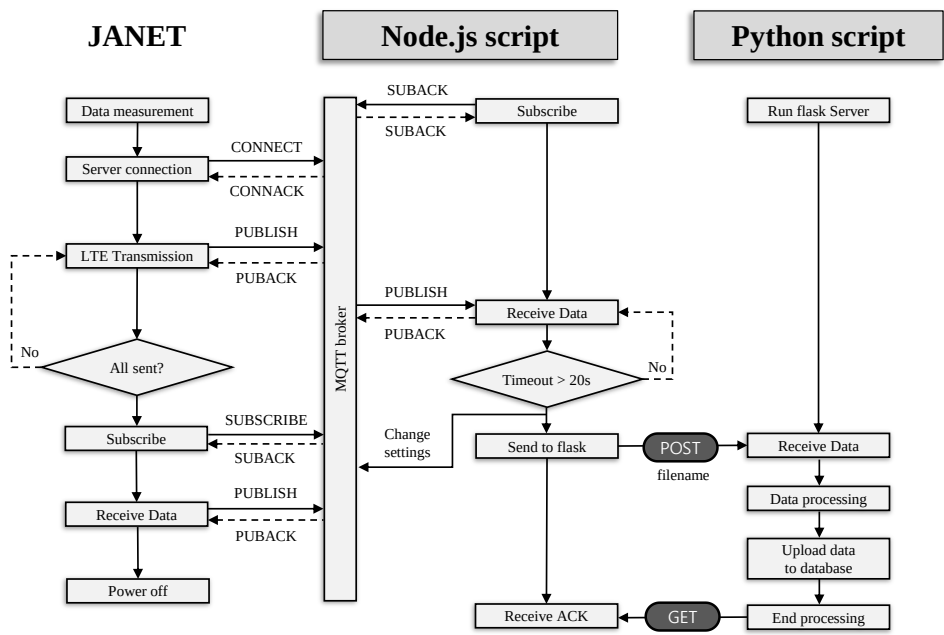


Figure 3-2: Script-based framework diagram

3.2 Data Schema

JANET can acquire data from three-axis accelerometers and up to eight-channel ADCs. However, in this study, only three strain gauges and one crack gauge are used to read data from a total of four ADC channels. Accordingly, the raw data sent to the server from the sensor is the index of the data, three-axis accelerations (acc_x, acc_y, acc_z), and ADC values from four channels ($str_1, str_2, str_3, crack$). Note that the battery voltage, LTE signal strength, data trigger type, and vibration trigger count are transmitted along with the measurement data (Figure 3-3). This is to monitor not only the bridge but also the status of the sensor.

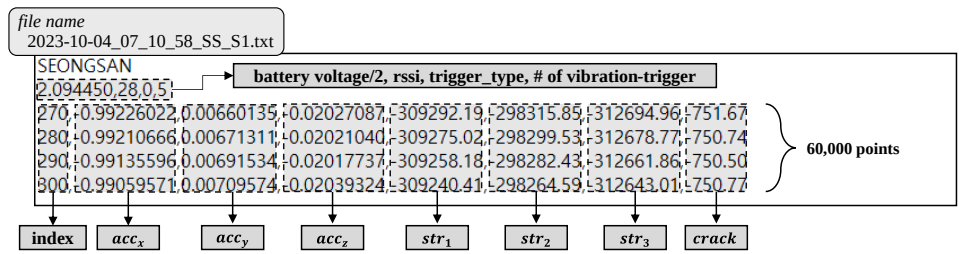


Figure 3-3: Raw data format from JANET

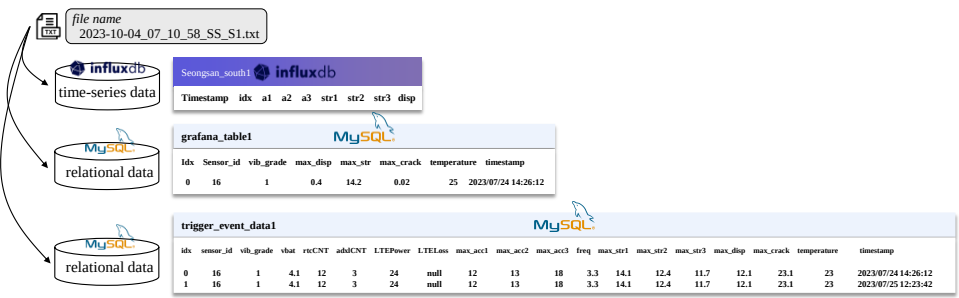


Figure 3-4: Raw data-based database configuration

The system utilizes cloud-based algorithms to process raw data from sensors and store derived values in two types of databases. The first database, InfluxDB, is ideal for time-series data, being schemaless and based on the NoSQL concept. InfluxDB records raw sensor measurements (acceleration, strain, crack) and cloud-estimated displacements. Also, users can set retention policies and periods to manage data accumulation, ensuring efficient use of cloud capacity in the future (Chapter 3.4). MySQL, a relational database, is employed to derive representative values from 60,000 data points per measurement. It comprises two tables: one for web dashboard visualization (Chapter 3.5) and another for detailed engineer analysis. The latter table generates various indicators from each measurement, facilitating long-term data analysis. Results from this analysis are presented in Chapter 4.

3.3 Strain Offset Compensation

In general, strain offset is calibrated by subtracting the average of the first strain value or the entire data range from the strain data [38]. However, this method is not accurate because it does not consider the vibration of the bridge. Fortunately, JANET is designed to measure acceleration and strain simultaneously, which allows us to approach this problem from an engineering perspective. Due to the nature of a bridge, all deformations that occur in the bridge are due to vehicle traffic, and vibrations occur when vehicles pass. Conversely, it can be inferred that the least vibration occurs when there is the least vehicle traffic, and the deformation caused by vehicles is also minimized. Based on this, the authors propose an algorithm to correct the origin of the strain of a bridge based on acceleration data. The details of the strain offset compensation method are shown in Algorithm 1, providing a detailed flow of strain calibration.

Algorithm 1: Strain offset compensation

Inputs: $\mathbf{acc}$ ($L \times 1$) - vertical acceleration data, $\mathbf{strain}$ ($L \times 3$) – strain data, N – length of the data

Output: calibrated strain ($L \times 3$)

Step 1: Define window size and step size

The window size N determines the width of the moving window that is used to calculate the standard deviation. The step size 1 indicates that the window is shifted by one sample at a time.

Step 2: Initialize variables

The $\mathbf{std}_{min}$ variable stores the minimum standard deviation that has been found so far. The $\mathbf{index}_{optimal}$ variable stores the index of the window with the minimum standard deviation.

Step 3: Calculate standard deviation for each window

For each index i , the standard deviation σ for the window $\mathbf{acc}[i:i+N]$ is calculated.

Step 4: Update minimum standard deviation and optimal index

If the standard deviation σ for the current window is smaller than the current $\mathbf{std}_{min}$, then the $\mathbf{std}_{min}$ and $\mathbf{index}_{optimal}$ are updated.

Step 5: Calculate optimal index

The optimal index is the index of the window with the minimum standard deviation.

Step 6: Calculate offset

The offset is calculated using the strain value at the optimal index.

Step 7: Calibrate strain data

The strain data is calibrated by subtracting the offset.

3.4 Displacement Estimation using Data Fusion

Instead of using a direct displacement measuring system, displacement could be estimated indirectly. Park et al. proposed the displacement estimation method using multimeric data fusion, which is obtained as:

$$\{u\}_{m \times 1} = \Phi_{m \times r} \{q\}_{r \times 1} \quad (12)$$

$$\{\varepsilon\}_{n \times 1} = \Psi_{n \times r} \{q\}_{r \times 1} \quad (13)$$

where $\{u\}_{m \times 1}$, $\{\varepsilon\}_{n \times 1}$, $\{\Phi\}_{m \times r}$ and $\{\Psi\}_{n \times r}$ are displacement, strain, displacement mode shape and strain mode shape, respectively. $\{q\}_{r \times 1}$ is the modal coordinate, and r is the number of used modes. By applying Moore-Penrose pseudo inverse (13) to (12),

$$\{u\}_{m \times 1} \cong \alpha \Phi_{m \times r} (\Psi_{n \times r})^\dagger \{\varepsilon\}_{n \times 1} \quad (14)$$

$$= \alpha D_{m \times n} \{\varepsilon\}_{n \times 1} \quad (15)$$

where $D_{m \times n}$ is a transformation matrix, α is the scaling factor. The relationship between displacement-strain for a simply supported beam is presented as:

$$\varepsilon_k(t) = -u''(x_k, t)y = -y \sum_j \phi''_{k_j} q_j(t) \quad (16)$$

$$= \frac{y\pi^2}{L^2} \sum_j j^2 \sin \frac{j\pi x_k}{L} q_j(t) \quad (17)$$

where u_i and ε_k are the respective displacement at x_i and strain at x_i , ϕ_{ij} is a mode shape value of the j th mode at x_i , y is the neutral axis of the beam, q_j is the j th modal coordinate, and L is the length of the beam.

Here, the generalized coordinate $q_j(t)$ can be expressed in terms of strain as:

$$\begin{bmatrix} q_1 \\ \vdots \\ q_r \end{bmatrix} = \frac{L^2}{y\pi^2} \begin{bmatrix} \sin \frac{\pi x_1}{L} & \cdots & r^2 \sin \frac{r\pi x_1}{L} \\ \vdots & \ddots & \vdots \\ \sin \frac{\pi x_n}{L} & \cdots & r_2 \sin \frac{r\pi x_n}{L} \end{bmatrix}^\dagger \begin{bmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_r \end{bmatrix} \quad (18)$$

Substituting (18) into (16)

$$D = \frac{L^2}{y\pi^2} \Phi \begin{bmatrix} \sin \frac{\pi x_1}{L} & \cdots & r^2 \sin \frac{r\pi x_1}{L} \\ \vdots & \ddots & \vdots \\ \sin \frac{\pi x_n}{L} & \cdots & r_2 \sin \frac{r\pi x_n}{L} \end{bmatrix}^\dagger \quad (19)$$

Then, a strain-based displacement with the same shape as the actual displacement was obtained. To scale the strain-based displacement to the actual displacement unit, it is required to be multiplied by α . Park et al. proposed its scaling factor as:

$$\alpha = \sqrt{\frac{S_{acc}^{disp}(f_n)}{S_{strain}^{disp}(f_n)}} \quad (20)$$

where S_{strain}^{disp} is the PSD of the strain-based displacement, S_{acc}^{disp} is the PSD of the acceleration-based displacement and f_n is the dominant natural frequency. Through this data fusion technique, JANET can estimate displacement from its measurement. In this study, displacement is estimated using one acceleration and three strain data.

3.5 Design of Dashboard for Data Visualization

Most IoT sensors simply store, or process measured data in a form that only engineers can understand. However, infrastructure facilities are owned by the state, and it is difficult for all government officials who make decisions about the safety of structures to perform data analysis themselves. Therefore, it is considered meaningful that IoT sensors not only continuously acquire data, but also analyze the collected data in real time after each measurement and present the results in a way that can be easily understood briefly. For this purpose, the results are shown using Grafana, a web-based visualization platform.

Grafana is a popular open-source platform for building interactive dashboards and visualizations. It allows users to connect to diverse data sources, including time-series databases, relational databases, and APIs, and then visualize the data in a variety of ways, such as graphs, charts, and maps. In this paper, the databases measured and collected by JANET are connected to Grafana to enable data-driven condition assessment in real time without the need for additional data processing. Figure 3-5 is a dashboard that visualizes the real-time monitoring results of the southern end of Seongsan Bridge. This dashboard receives the data measured by the actual sensor from the cloud server, stores the analysis results from the received raw data through the built-in cloud calculation in the database, and connects the database for Grafana visualization and each panel of Grafana.

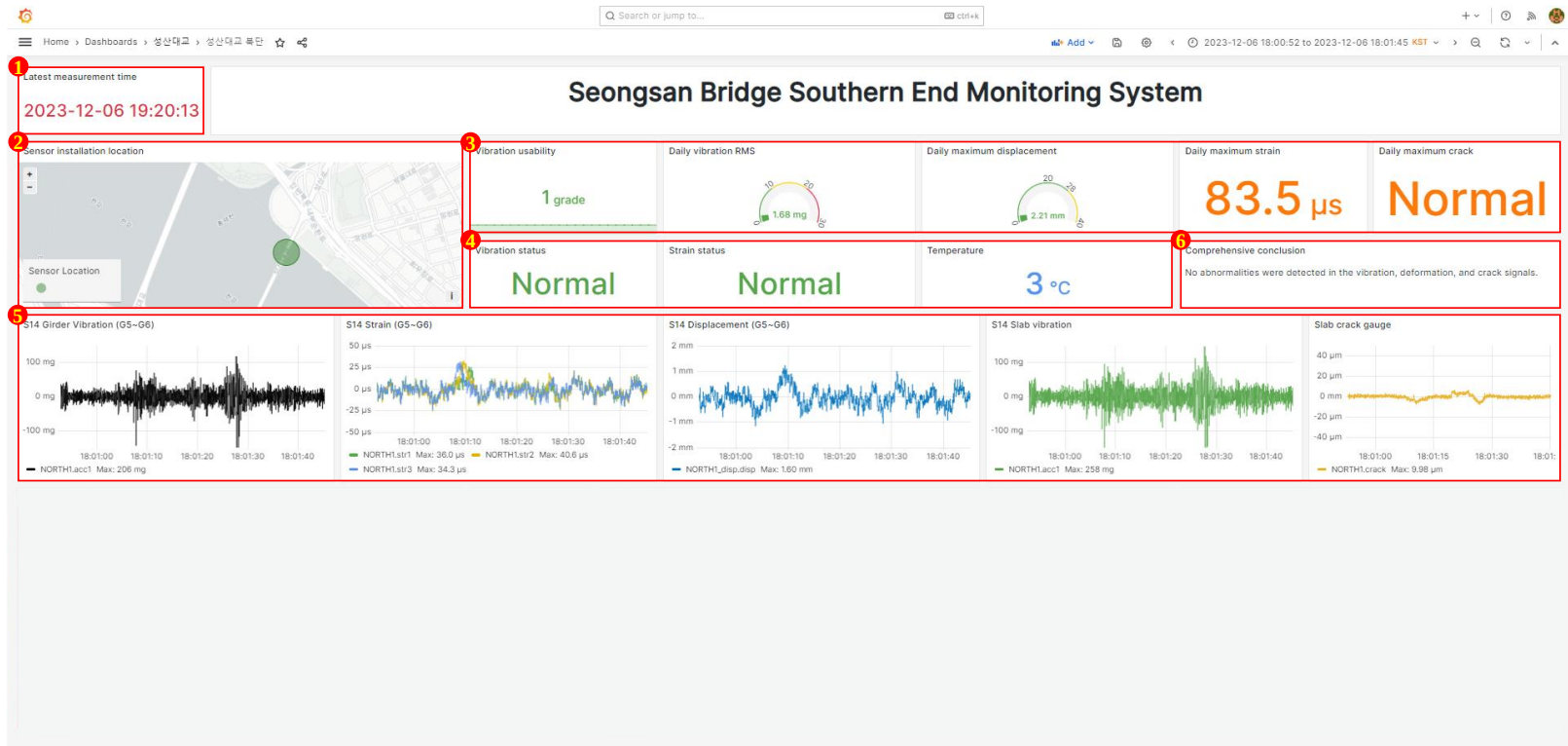


Figure 3-5: Web-based real time Grafana dashboard

- (1) Latest measurement time: This panel displays the time when the sensor's data is received. This information is useful for checking if the sensor is operating normally and identifying potential problems.
- (2) Sensor installation location: This panel indicates the location where the sensor is installed on the bridge. It helps identify the specific area of the bridge being monitored.
- (3) Primary index: This panel exhibits basic physical indicators such as acceleration RMS and maximum displacement. These indicators are valuable for tracking the overall condition of the bridge.
- (4) Data-driven bridge assessment: This panel showcases the time-series data collected by the sensor, allowing tracking of the historical condition of the bridge.
- (5) Acquired data: This panel presents the time-series data collected by the sensor, providing insights into the historical condition of the bridge.
- (6) Comprehensive conclusion: This panel is designated for recording any additional information relevant to the dashboard.

4 Application Long-term Bridge Monitoring

This chapter introduces the application of JANET to a real bridge. The JANET system was installed on the Seongsan Bridge in Seoul, South Korea, in July 2023 and has been monitored for 4 months to date. The background of sensor installation, the installation process, single measurement data, and long-term data analysis are introduced.

4.1 Description of the Testbed



Figure 4-1: Seongsan Bridge, Seoul, South Korea

The Seongsan Bridge, a special bridge connecting Mapo-gu and Yeongdeungpo-gu in Seoul, is an aging structure that has been in operation for over 40 years. It carries an average of over 160,000 vehicles per day.

The bridge is made of steel and concrete, and monitoring its condition is crucial to ensure safety.

Despite a performance improvement project from 2017 to 2021, cracks were again found in the deck. This was due to several factors, including subcontracting of the repair work by the maintenance company to unlicensed companies, improper transportation of the deck, and the preparation of reports that were not relevant to the inspection results.

As a result, the Seoul Metropolitan Government decided to introduce the "Han River Bridge Online Safety Monitoring System" to ensure the long-term observation of the bridge. This system includes the JANET system proposed in this study.

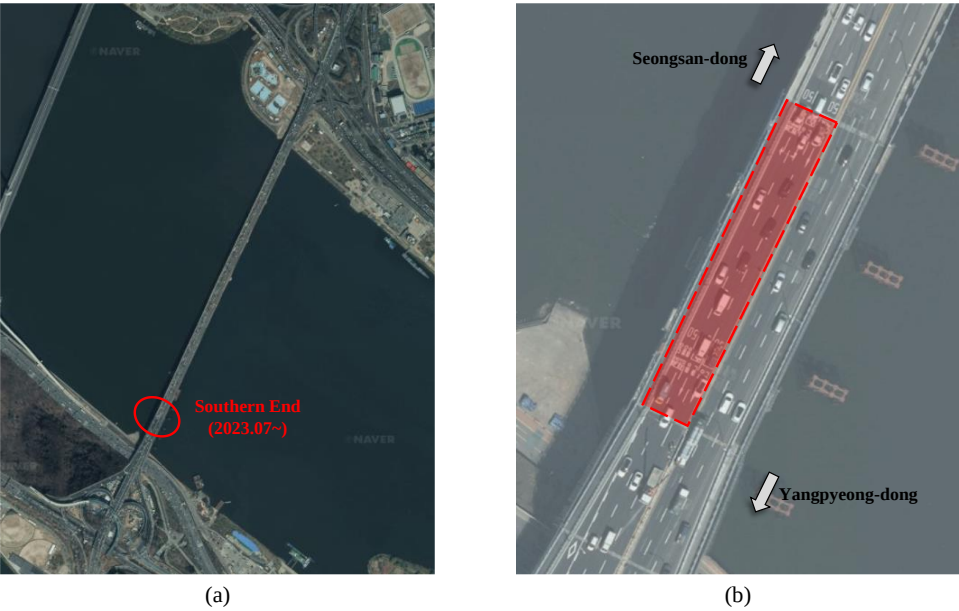


Figure 4-2: Satellite images for sensor deployment location

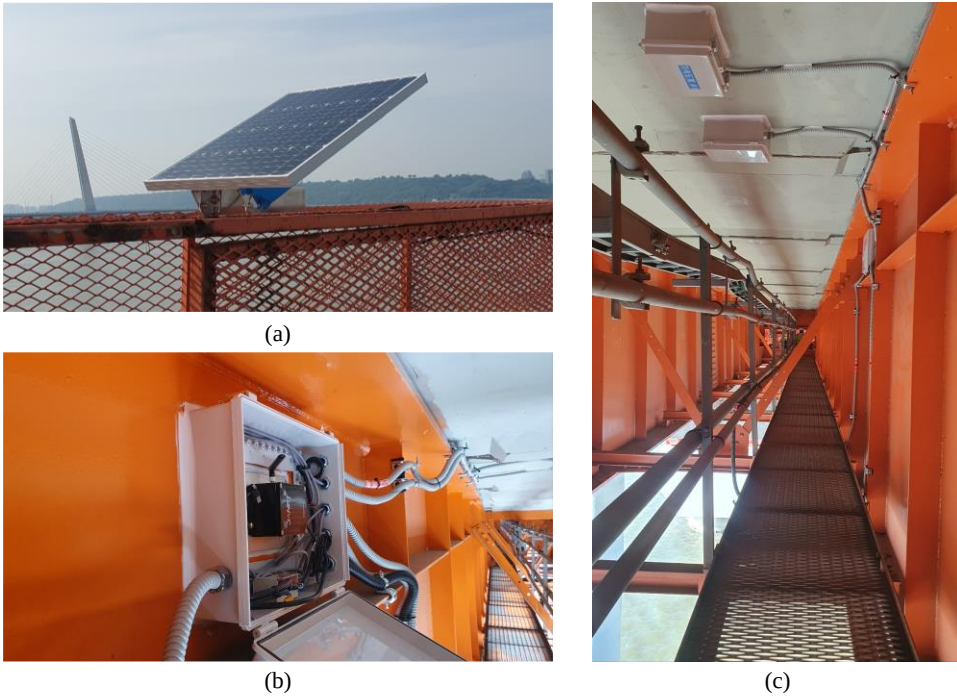
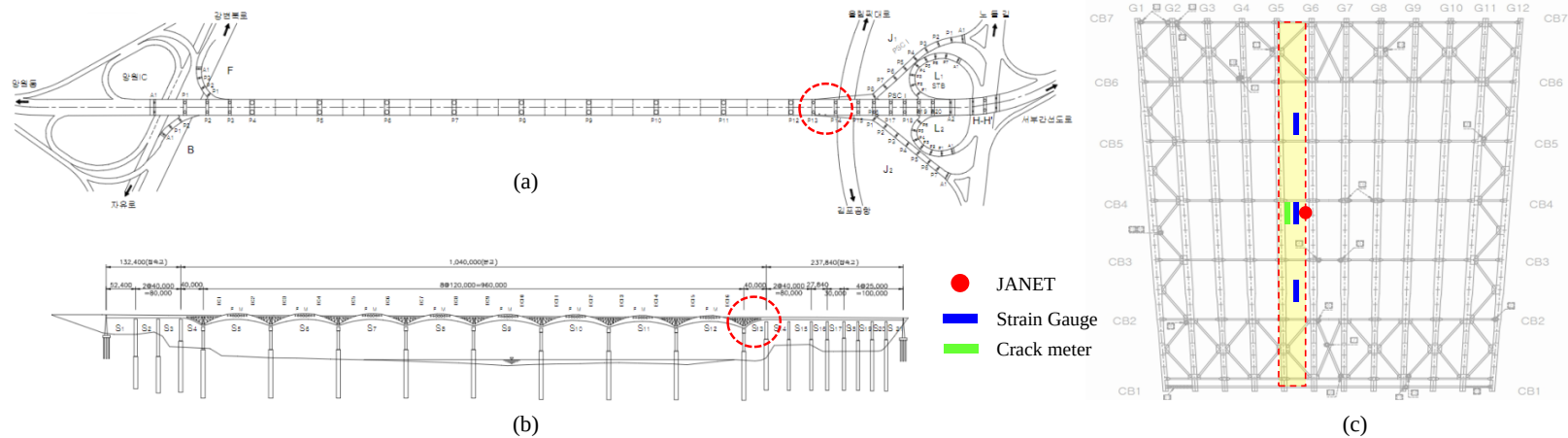


Figure 4-3: On-site images of Seongsan Bridge installation: (a) solar panel (b) sensor system (c) overview

In this study, sensors were applied to the S13 span of the Seongsan Bridge on July 6, 2023. A solar-charging system, as described in Chapter 2.5, was installed above the bridge inspection entrance to enable long-term operation without battery replacement, as shown in Figure 4-3 (a). Figures 4-3 (b) and (c) show the overall view of the installed sensors. The system installation was carried out over two days, from July 6 to 7.

The S13 span is a 34.2-meter span with 12 girders. This system installed one JANET, three welded strain gauges, and one crack meter in the CB2 to CB6 sections of the G5 to G6 girders, as shown in Figure 4-4 (c).



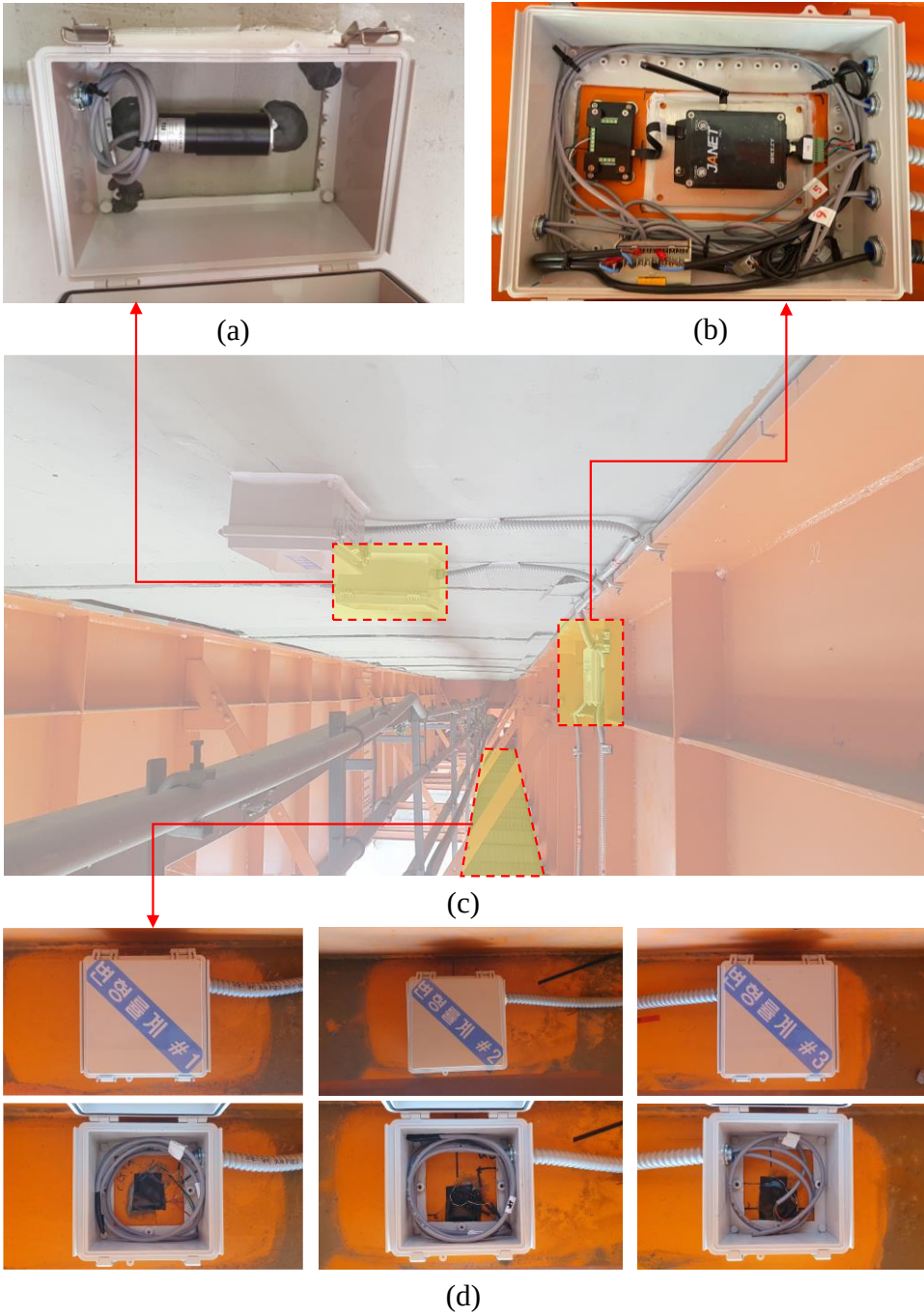


Figure 4-5: Installed sensors: (a) crack gauge (b) packaged JANET (c) system overview (d) strain gauges

4.2 Analysis on Data Collection

Ideally, the JANET system is designed to collect data 24 times a day through a timer-based trigger and up to 5 times a day through a vibration-based trigger. However, during the development and field application of the system, unexpected problems often occurred. The sensors were unable to charge properly due to cloudy weather throughout the summer, and, in some cases, the sensors were damaged due to overvoltage caused by lightning. Additionally, most of the data was lost due to insufficient debugging before applying software upgrades to the sensors and cloud servers. As a result of these trials and errors, the sensors and server code have now been stabilized, and the system has been receiving data without any loss for over a month.

2023

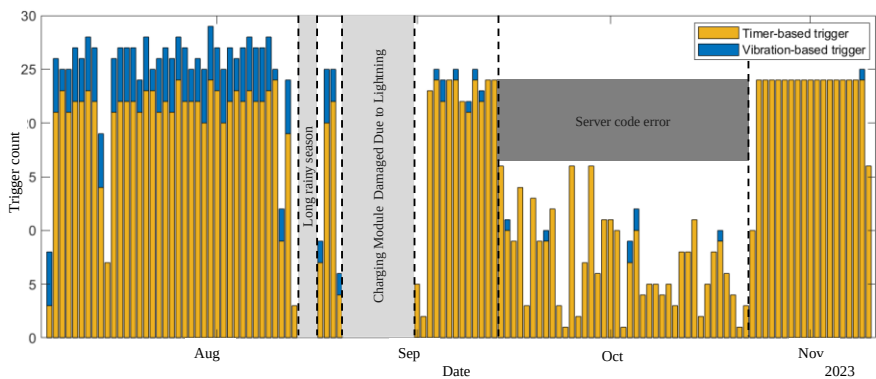


Figure 4-6: Data acquisition history

Figure 4-7 shows the time series data generated each time JANET is triggered. For visualization purposes, only the last 5 minutes of the 10-minute data are plotted.

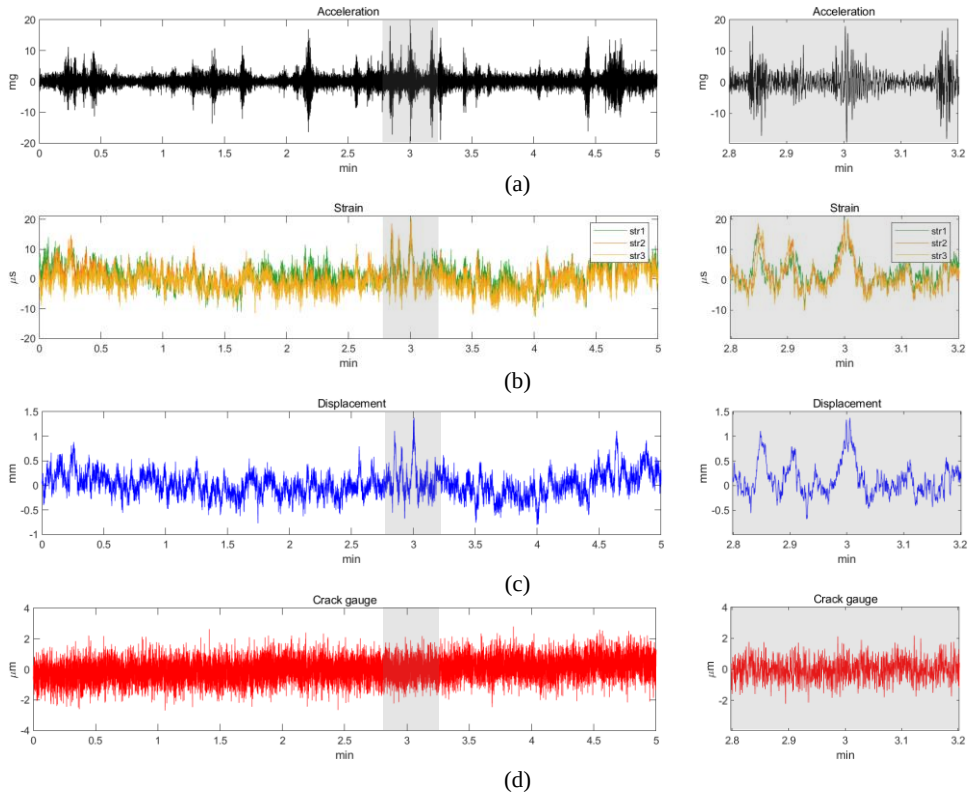


Figure 4-7: Raw data: (a) acceleration (b) strain (c) displacement (d) crack gauge

4.3 Analysis on Frequency

Frequency analysis of vertical accelerations measured by JANET revealed two distinct peaks at 3.271 Hz and 5.176 Hz. The two peaks were observed throughout the entire data range, suggesting that both frequencies could be associated with the girder. To resolve the ambiguity, the natural frequency of the adjacent ramp bridge was considered. The ramp bridge had a theoretical natural frequency of 5.663 Hz, which was close to the 5.176 Hz peak detected in the JANET data. It was, therefore, concluded that the 5.176 Hz peak was due to the vibration of the ramp bridge, and the 3.271 Hz peak was assigned as the natural frequency of the girder. This approach was effective in resolving the ambiguity and provided a more accurate determination of the girder's natural frequency.

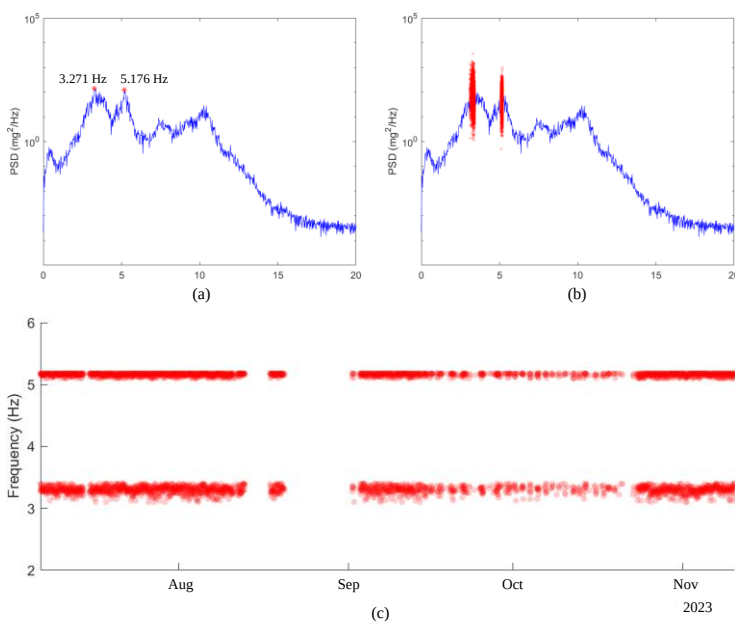


Figure 4-8: Frequency peak analysis: (a) single peak (b) multi peaks (c) peak history by date

4.4 Analysis on Displacement

The maximum displacement that occurred for each measurement was analyzed from data measured over 4 months. Figure 4-9 shows a histogram of the maximum deflection that occurred over 4 months. This analysis reveals that long-term monitoring using JANET detected a maximum deflection of up to 2.61 mm. The distribution of the obtained data was analyzed using a Gaussian distribution.

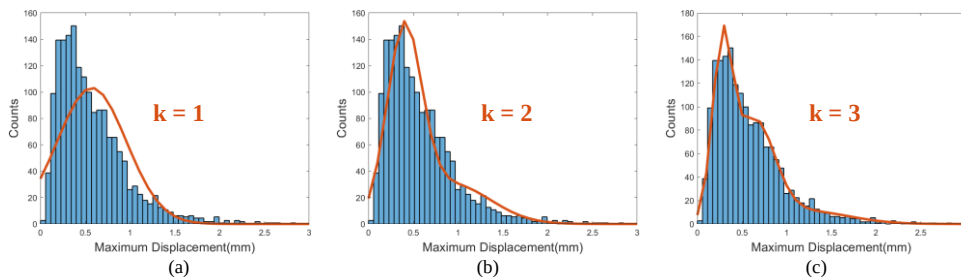


Figure 4-9: Data distribution of maximum displacement with parameter change: (a) $k=1$ (b) $k=2$ (c) $k=3$

Since the data is evenly distributed over a sufficient period, it will match the Gaussian distribution if the type of vehicle is not biased. Otherwise, it will exhibit a different pattern. The analysis confirmed that the Gaussian distribution with $k=1$ did not match the actual data. On the other hand, the higher the value of k ($k=2$ or $k=3$), the better the data distribution could be explained. This suggests that vehicles passing on Seongsan Bridge can be roughly divided into 2 to 3 types.

Table 4-1 presents the maximum allowable bridge deflection criteria from the Highway Bridge Design Standard (Limit State Design),

established by the Ministry of Land, Infrastructure and Transport [39]. Utilizing the most stringent standard, defined as $L/1000$, the calculated maximum allowable deflection for the section where JANET is installed is 37.6mm ($L/1000 = 37.6\text{m}/1000$). This value is significantly lower than the observed maximum displacement of 2.61 mm over the 4 month monitoring period. Consequently, it is evident that the bridge's behavior remained well within the acceptable range, as demonstrated by the extensive long-term data collection.

Table 4-1: Deflection criteria from the Highway Bridge Design Standard

-	Criteria
Vehicle Load, general	L/800
Vehicle load, pedestrian load	L/1000
Vehicle load on cantilever	L/300
Vehicle load or pedestrian load on cantilever	L/375

4.5 Analysis on Crack

Seongsan Bridge underwent performance improvement work, yet numerous cracks necessitated further repair. Since sensor installation required completed repairs, the JANET system was installed in July 2023 when the southern end of the bridge repairs was finished, revealing no visible cracks. Paint covered the entire surface, further concealing any repair traces. For this reason, the crack gauge was attached to the center of the span, which was believed to receive a relatively high load. Over three months, the distribution of maximum crack values measured by the crack system was analyzed. The resulting graph confirmed that the larger k becomes, the more accurate it becomes.

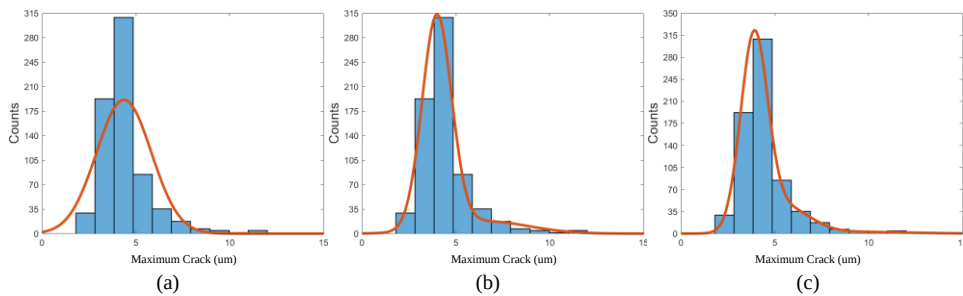


Figure 4-10: Data distribution of maximum crack width with parameter change: (a) $k=1$ (b) $k=2$ (c) $k=3$

The Ministry of Land, Infrastructure, and Transport defines the allowable crack width for reinforced concrete structures in wet environments like Seongsan Bridge as 0.3mm through the KDS 14 20 30 concrete structure design standards [40]. The maximum crack value

measured by JANET at each trigger was 66.8 μm , significantly lower than the allowable limit.

Table 4-2: The allowable crack width of reinforced concrete structures

Steel type	Environmental conditions for corrosion of steel materials			
	Dry environment	Wet environment	Corrosive environment	Highly corrosive environment
Reinforcing steel	Greater of 0.4mm and $0.006C_c$	Greater of 0.3mm and $0.005C_c$	Greater of 0.3mm and $0.004C_c$	Greater of 0.3mm and $0.0035C_c$
Prestressed steel	Greater of 0.2mm and $0.005C_c$	Greater of 0.2mm and $0.004C_c$	-	-
C_c : Minimum concrete cover thickness (mm) between the surface of the outermost reinforcing steel and the concrete surface				

4.6 Discussions

The JANET system, a bridge monitoring sensor, is capable of simultaneously measuring 3-axis acceleration and up to 8 channels of analog sensors at 100Hz. Equipped with an EDS circuit, it allows for adjustable data collection, catering to various purposes such as vibration-based or timer-based triggers. We have developed a cloud-based server that wirelessly acquires data through an LTE modem, incorporating a framework for receiving, storing, and executing cloud operations on the collected data. Additionally, a platform links preprocessed data to a visualization dashboard, enabling facility safety managers to make informed decisions without requiring an engineering background. This system has been implemented on the Seongsan Bridge in Seoul and monitored for more than 4 months. Notably, the calculation of the maximum allowable deflection of the Seongsan Bridge, following the Highway Bridge Design Standard, yields a value significantly higher than the observed maximum displacement over three months, affirming the bridge's compliance within acceptable operational parameters. Moreover, the scrutiny of crack data unveils that the measured crack width remains well below the allowable limit, as stipulated by pertinent standards. This emphasizes the system's efficacy in monitoring and evaluating structural integrity.

5 Conclusion

In this paper, an IoT sensor, JANET, and cloud computing are proposed for the long-term monitoring of bridges. JANET is capable of simultaneously measuring acceleration and strain, and its precision has been verified with RMS errors of 0.1 mg and 1.52 μ s, respectively. Furthermore, a cloud framework for collecting and processing data has been developed. Time-series databases, such as InfluxDB, and relational databases, such as MySQL, were established within the cloud, with scalable designs for each database schema. The developed system was installed at the southern end of Seongsan Bridge, Seoul, and its usability was validated through application from July 2023 for a period of four months. After each data measurement, successful extraction of vibration, strain, natural frequencies, and displacement was confirmed.

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국문초록

박준영

토목공학과 구조공학 전공

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교량은 주요한 사회기반시설물로서 정기적인 정밀점검을 통해 그 안전성을 담보한다. 하지만 국내의 공용년수 30 년 이상의 노후 교량은 2020 년 17.9%에서 2030 년 49.7%로 10 년간 급격히 증가할 전망이며, 줄어드는 건설 인력으로 인해 인력 중심의 기존 점검방식은 급격히 늘어나는 노후 교량을 관리하는데 한계가 있다. 더욱이 교량의 안전점검 시 실시하는 내하력 평가에서는 교량의 활하중 저항능력을 나타내는 공용내하력을 산정하기 위해 변위를 계측해야하나, 종래의 변위계측 방식 (LDV, LVDT)으로는 하천 교량이나 장대 교량에 계측기를 설치할 수 없어 제대로 된 점검이 어려운 실정이다. 이러한 문제를 극복하기 위해 본 논문에서는 IoT 센서 (JANET)와 클라우드 기반의 교량 모니터링 시스템을 개발하였다. JANET 은 3 축 가속도와 최대 8 채널 변형률을 동시에 계측할 수 있으며, EDS 기반 트리거

회로를 통해 저전력 상태를 유지하다 일정 시각이나 과도한 진동이 발생할 경우 트리거되어 데이터를 획득한다. 또한, LTE Cat 1 모뎀을 내장하여 계측한 데이터를 무선으로 클라우드 서버에 전송한다. 클라우드 서버는 데이터 획득을 위한 node.js 기반 스크립트와 데이터 처리를 위한 python 기반 스크립트로 구성되어 있으며, 변형률 원점 보정 알고리즘, 가속도-변형률 기반 변위 추정 알고리즘이 내장되어 계측 데이터로부터 교량의 주요 지표를 실시간으로 산출한다. 산출 데이터는 Influxdb, mysql 과 같은 데이터베이스에 저장하고, 각 데이터베이스는 웹 기반 대시보드 Grafana 와 연동되어 교량의 관리 주체가 별도의 배경지식 없이도 데이터를 기반으로 의사결정을 내릴 수 있도록 돕는다. 본 개발 시스템은 2023 년 7 월부터 서울에 위치한 성산대교 남단부에 설치되어 현재까지 계측을 수행하고 있으며, 계측 기간 중 발생한 최대 변위 (2.61 mm)와 최대 균열 (66.8 μ m)이 센서 부착 지점의 설계기준 대비 6.94%, 22%로서 안정적으로 거동함을 확인하였다.